ii) *Cut-off region*

A small collector current exists even after the base current is reduced to zero. This region below the curve for $I_B = 0$ is called cut-off region because the main collector current is cut-off.

iii) *Active region*

The central region of the curves is called active region. In this region, the base-emitter junction is forward biased and the collector-base junction is reverse biased. The transistor in this region can be used for voltage, current and power amplification.

iv) *Breakdown region*

If the collector-emitter voltage is increased beyond the rated value given by the manufacturer, the collector current increases enormously leading to the junction breakdown of the transistor. This avalanche breakdown can damage the transistor.

Output impedance

The ratio of the change in the collector-emitter voltage (ΔV_{CE}) to the corresponding change in the collector current (ΔI_C) at constant base current (I_B) is called output impedance (r_0).

$$r_o = \left(\frac{\Delta V_{CE}}{\Delta I_C}\right)_{I_B} \qquad (10.4)$$

The output impedance for transistor in common emitter configuration is very low.

iii) Current transfer characteristics

This gives the relationship between the collector current (I_C) and the base current (I_B) at constant collector-emitter voltage (V_{CE}) and is shown in Figure 10.33. It is seen that a small I_C flows even when I_B is zero. This current is called the common emitter leakage current (I_{CEO}), which is due to the flow of minority charge carriers.

Forward current gain

The ratio of the change in collector current (ΔI_C) to the corresponding change in base current (ΔI_B) at constant collector-emitter voltage (V_{CE}) is called forward current gain (β).

$$\beta = \left(\frac{\Delta I_C}{\Delta I_B}\right)_{V_{CE}} \qquad (10.5)$$

Its value is very high and it generally ranges from 50 to 200.

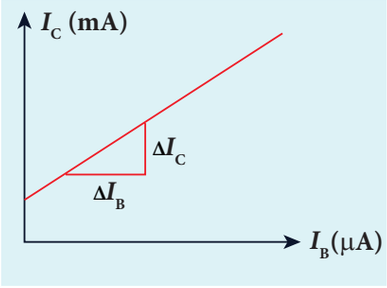


Figure 10.33 Current transfer characteristics

10.4.3 Relation between α and β

There is a relation between current gain in the common base configuration α and current gain in the common emitter configuration β which is given below.

$$\alpha = \frac{\beta}{1+\beta} \text{ (or) } \beta = \frac{\alpha}{1-\alpha} \qquad (10.6)$$

Note

The collector current is independent of the collector-emitter voltage in the active region.

EXAMPLE 10.6

In the circuit shown in the figure, the input voltage V_i is 20 V, $V_{BE} = 0$ V and $V_{CE} = 0$ V. What are the values of I_B, I_C, β?

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